

MyDrinkShop – Software Requirements Specification

Version: 2.0 **Date:** 09.03.2026 **Status:** Updated & Corrected (Lab01 Inspection)

1. Introduction

1.1 Purpose

This document describes the functional and non-functional requirements for the **MyDrinkShop** application — a JavaFX desktop point-of-sale (POS) management system for a drink shop.

1.2 Scope

MyDrinkShop supports management of products (drinks), orders, recipes (retete), and stock (stocuri). It provides a GUI for daily shop operations, receipt generation, CSV export, and daily revenue reporting.

1.3 Definitions

Term	Definition
Product	A drink available for sale in the shop
Order	A customer purchase consisting of one or more order items
OrderItem	A single line in an order: one product with a quantity
Reteta	A recipe describing ingredients and quantities needed to prepare a product
Stoc	Current inventory quantity of a raw ingredient
CategorieBautura	Drink category (e.g., CLASSIC_COFFEE, TEA, JUICE)
TipBautura	Drink variant (e.g., DAIRY, PLANT_BASED, WATER_BASED)

2. Functional Requirements

2.1 Product Management

FR-P1: Add product: unique positive ID, non-empty name, strictly positive price, valid category and type (not ALL).

FR-P2: Update an existing product's name, price, category, and type.

FR-P3: Delete a product by ID.

FR-P4: Display all products in a table view.

FR-P5: Filter products by category or type. ALL returns all products.

FR-P6: A product may only be added if a recipe with the same ID exists and no product with that ID already exists. *[Added v2.0]*

2.2 Order Management

FR-O1: Create a new order by selecting products and quantities.

FR-O2: Validate before saving: at least one item, valid product ID, positive quantity, non-negative total.

FR-O3: Compute total price: sum of (price × quantity) for each item. Must be called before saving.
[Clarified v2.0]

FR-O4: Finalize (save) the current order and generate a receipt.

FR-O5: After finalizing, clear the current order and start a new one with the next ID.

FR-O6: Display all finalized orders.

2.3 Recipe (Reteta) Management

FR-R1: Add a new recipe with a list of ingredients (name + positive quantity).

FR-R2: Recipe ID equals the corresponding Product ID. *[Clarified v2.0]*

FR-R3: Update or delete a recipe.

FR-R4: Display all recipes showing ingredients and quantities.

2.4 Stock (Stoc) Management

FR-S1: Track available quantities of raw ingredients.

FR-S2: Before processing a product, check sufficient stock for all recipe ingredients.

FR-S3: If sufficient, deduct ingredient quantities from stock.

FR-S4: If insufficient, throw `IllegalArgumentException` with a descriptive message.

FR-S5: Stock quantities are stored as **double** values (fractional amounts supported). *[Corrected v2.0]*

2.5 Receipt Generation

FR-RG1: Generate a text receipt showing Order ID, per-item details (name, price, qty, line total), and grand total in RON.

2.6 CSV Export

FR-E1: Export all orders to CSV: columns OrderId, Product, Quantity, Price.

FR-E2: Include per-order totals and daily grand total with current date.

2.7 Daily Revenue Report

FR-DR1: Display total daily revenue (sum of all order totals).

FR-DR2: Provide count of total orders.

3. Non-Functional Requirements

ID	Requirement
NFR-1 (Persistence)	All data persisted to text files. Changes written immediately after each modification.
NFR-2 (Validation)	All input validated before saving. Validators must be invoked in the service layer. [Corrected v2.0]
NFR-3 (Architecture)	Layered architecture: UI → Service → Repository → File.
NFR-4 (Technology)	Java 17+, JavaFX, Maven.
NFR-5 (Usability)	Error dialogs shown for invalid input or failed operations.
NFR-6 (Correctness)	No duplicate methods. Order exposes a single getTotalPrice(). [Added v2.0]

4. Constraints

- Single-user desktop application (no network or multi-user support).
- File paths for data files are relative to the project root (e.g., data/products.txt).
- The ALL value in `CategorieBautura` and `TipBautura` is a UI filter sentinel only and must not be assigned to actual products.

5. Changes from v1.0 to v2.0

#	Change	Reason
1	FR-P1: ID > 0, price > 0 made explicit	Ambiguity in original
2	FR-P6 added: product needs pre-existing recipe	Missing constraint in v1.0
3	FR-R2 added: Reteta ID = Product ID	Undocumented coupling in v1.0
4	FR-S5 added: stock quantities are double	Code bug (int cast) fixed
5	NFR-2 corrected: validators called in service layer	Validators existed but not invoked
6	NFR-6 added: no duplicate methods	getTotal() removed from Order
7	Constraint: ALL enum values are filter sentinels	Ambiguity in original

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